

Effects of Innovative MWFs on Sustainability and Performance of Industrial Machining and Grinding Processes

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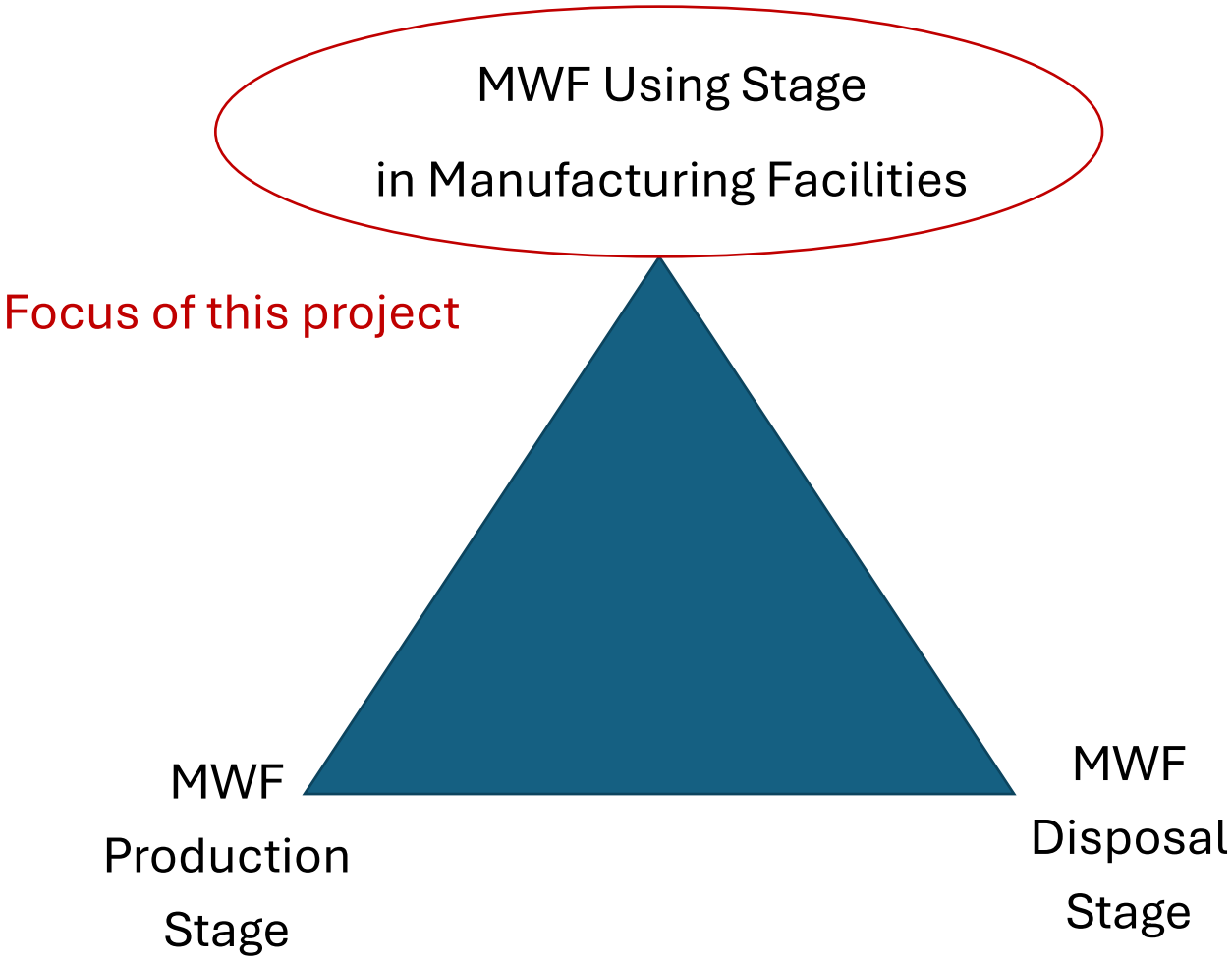
1.Introduction:

Metalworking Fluids (MWFs) play an essential role by reducing heat and friction increasing tool life and workpiece quality. These fluids are typically complex mixtures of oils, emulsifiers and other chemical agents to improve machining efficiency and surface quality. However, most MWFs are petroleum based, raising significant environmental concerns due to their dependency on fossil fuels resulting in greenhouse gases. Bio based MWFs have emerged as promising alternatives, offering potential benefits by reducing carbon footprint and enhancing machining process. Bio based MWFs can be formulated with low oil content decreasing fossil fuel dependency. Research has shown that bio based MWFs improve grinding efficiency, reduce energy consumption and higher surface quality.

Evaluation of a water based MWF (I-MWF) for grinding AISI 52100 Steel commonly used in applications requiring in high strength and wear resistance. We investigate the I-MWFs impact on carbon footprint and grinding performance, comparing its efficiency, surface quality and sustainability against water based MWF (B-MWF). By evaluating the I-MWFs on specific grinding energy (SGE), Friction Factor (FF) and surface Roughness parameters. The project is to show the benefits of Bio based MWFs in industrial use [3].

2.Problem Statement:

The use of metalworking fluids (MWFs) in machining and grinding process has significant impact for both environment and industrial efficiency. Traditional MWFs are related with high carbon emissions, especially during production and disposal. It causes various health and environmental hazards. In this project we evaluate MEFs to determine their potential in reducing the carbon footprint and improving operational performance of the machining. In this report we identify different fluids that will improve machining process but also supports sustainable manufacturing practices[1].



3.Experimental Setup:

3.1 CNC

To perform this experiment, we used a highly repeatable, precise and adjustable setup to investigate different grinding parameters. A CNC mill satisfies these requirements and simplifies the test set up due to its adaptability. The CNC mill used in this experiment was HAAS VF-2 which is five axis machine. The maximum spindle speed is 8100rpm.



Figure 1: HAAS VF-2

3.2 Wheel (Arbor)

To scratch the surface of the AISI 52100 Steel we used a diamond girt of size 230/270. This allowed for a more controlled data collection and limited amount of analysis was required.



Figure 2: Arbor with Diamond Grit

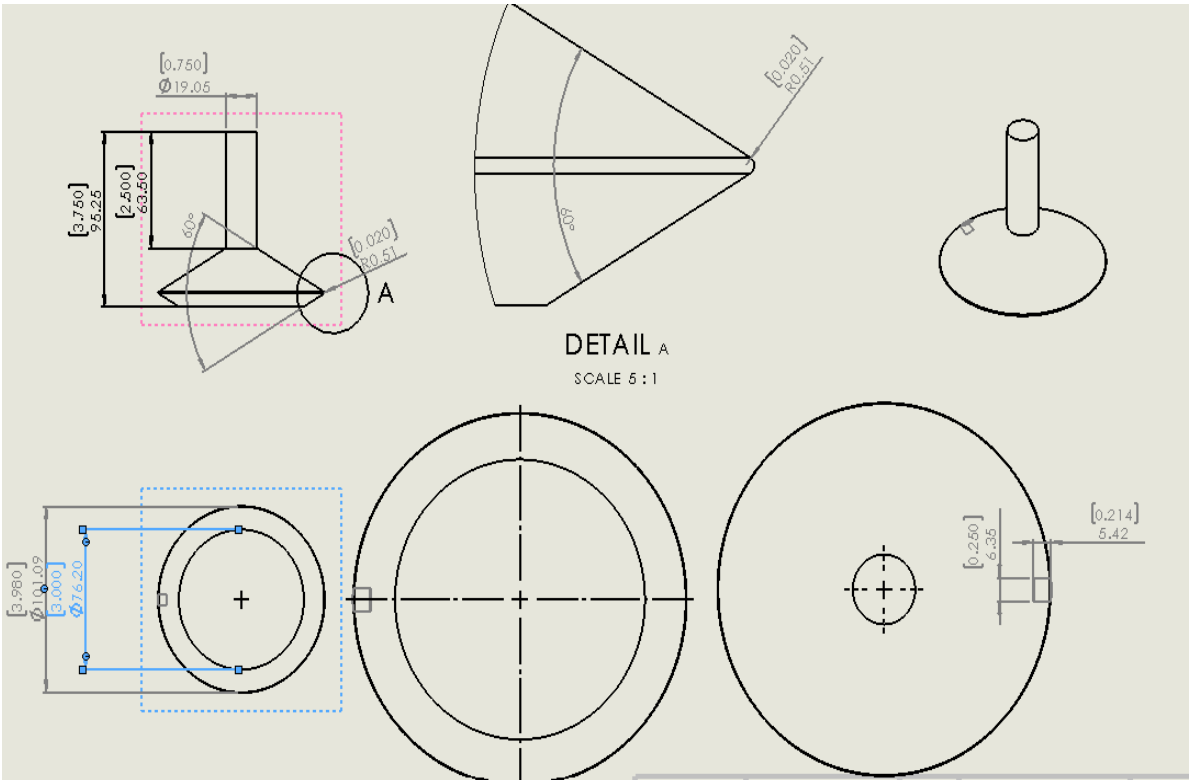


Figure 3: Drawing of the arbor

3.3 Force Sensor:

To record the force data, we used a sensor to capture the data. For recording these forces, the AISI 52100 Steel was fixed to the force sensor. The force sensor we used was Kistler 9347C which is used for its high sensitivity and broad range, The Kistler 9347C was used simultaneously. The amplifier transmitted all the signals to Ni 9223 logic controller. The controller maximum speed is 1MS/s.



Figure 4: Kistler Amplifier 5073B



Figure 5: Logic Controller 9223



Figure 6: Kistler 9347c

3.4 Fluid Delivery System:

The fluid delivery system is essential for providing consistent lubrication to the cutting zone during machining and griding process. It is designed to ensure that each MWF is delivered with precision, for comparison of the characteristics. The system has a nozzle of 1 mm diameter, delivering the fluid at a calibrated flow rate of 25ml/s and pressure of 4.3 bar. With this setup the fluid may effectively reach the cutting surface effectively. The jet velocity at the nozzle exit is 29.3m/s. The parameters minimize variability and provide consistent fluid delivering to the machine when the process is started. This system is very important to assess the environmental and operation of innovative MWFs to improve sustainability and efficiency in industrial machining.

- The parameters taken in the fluid delivery system are:
  If V_s=36.6, D_n=1 mm, and D_t=2.6 in, then,
  Q=25 ml/s
  P_t=4.3 bar
  V_t= 0.72 m/s
  V_n= 29.3 m/s

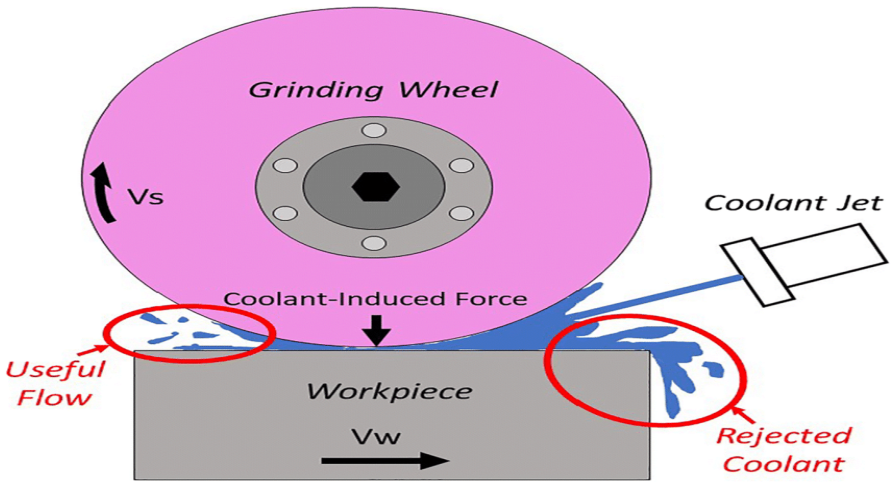


Figure 7: Fluid Delivery System

4. Testing:

With the data and test apparatus complete, testing is begun. The AISI 52100 steel gets bolted to the force sensor which gets bolted to L-shaped bracket. The bracket is secured in the CNC's and sample is oriented, so the edge of the spindle is normal to sample's surface. After this we then start making the coolant and fluid delivery system. The position of the spindle to the sample be manipulated as desired.

The fluid we used in this experiment was **HOCUT 795-RHS** and RO water. We used two 2 concentration of the liquid that for one experiment we used 16% of the **HOCUT 795-RHS** and we mixed with RO water. The other concentration we used was 5% of the **HOCUT 795-RHS** and then rest of them was mixed with RO water. Below Figure 8 is the way we used for mixing of the fluid.

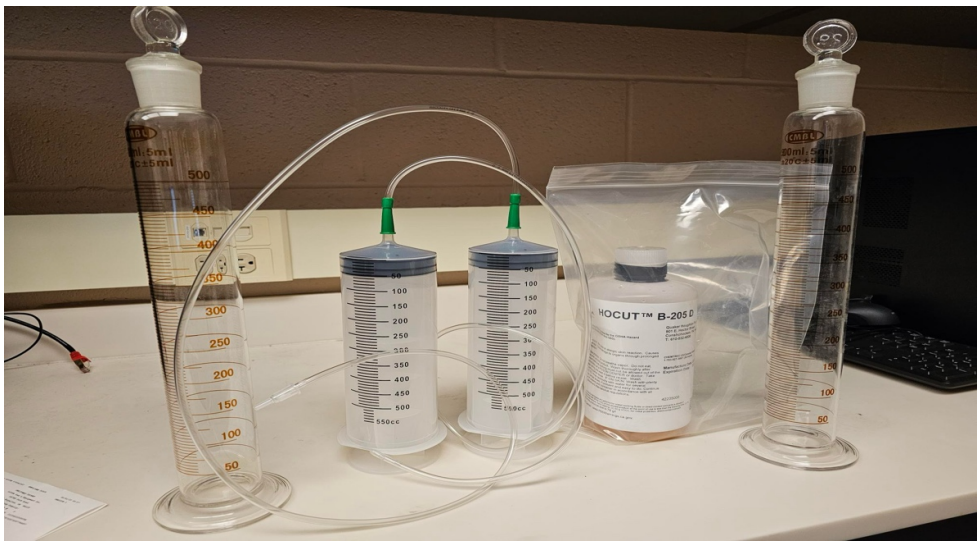


Figure 8- Fluid Delivery System.

We used AISI 52100 steel which are heated treated. We got this steel from Timken Company. We did a hardness test on the samples which were given to us. The average Hardness value we got was **904 Vickers.** Below are the Figure 9 and Figure 10 which shows the sample and the graph of the hardness test.

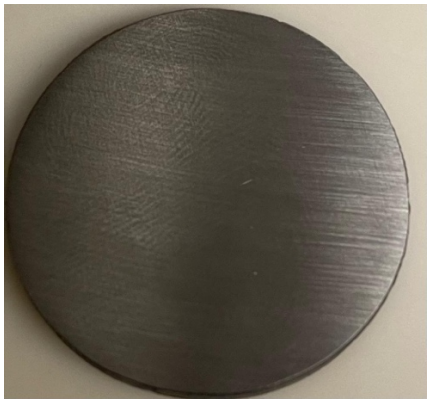


Figure 9 – AISI 52100 Steel with Heat Treated

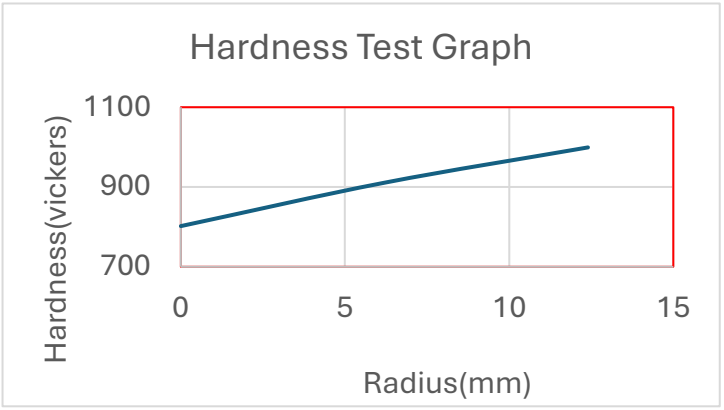


Figure 10 – Hardness graph of the AISI 52100 Steel heat treated.

For this test, it we used a shallow depth of cut. For doing this, the spindle was not rotated, and it was touched off with the help of the force sensor. As the spindle moved on the sample, if a force was observed it is said to be making a minor contact with the sample. Then the position of sample was known and could enter the CNC program.

When the testing begun, we also switch on the fluid delivery system and give it 2 sec to reach the sample and then we turn on the CNC program. The tips of the fluid system are kept aligning to the sample. The program was developed to the scratch the surface from one side to another side. The feed rate and spindle rate are adjusted according to the testing parameters. The program was written in way that it is stopped after each pass. This allowed for time in between to process data and ensure that everything was reading correctly. The scratches were done in the feed direction of the spindle rotation.

We made 5 scratches for the concentration of 16% fluid and 5 scratches for the 5% fluid. We made sure for every 5 scratches we changed the arbor to get the result as we needed. The girt size we used was Diamon arbor of 230/270. The spindle speed was 4775 rpm and feed was 6m/min for all the 10 scratches. The other parameters were kept constant. Below is the Figure 11 where we can see the scratches for different fluids.

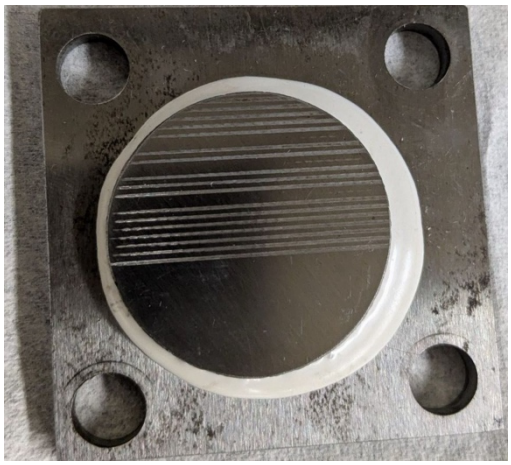


Figure 11- Test done on the sample

The most important parameter we consider here is the depth of the scratch. As we can observe when the spindle moved from right to left, the depth of scratch decreased until it failed to remove any material. The average depth of cut for 16% is 10.95µm and for 5% the average depth of cut was 23.17µm.

After the scratches were made, we need more information to create correlation and draw conclusions from the sample we had. The data can be obtained using Keyence microscope which is shown in Figure 12. Using the surface roughness feature we were able to gather 4 parameters in two directions of grinding.

1. Normal to grinding
2. In direction of grinding.

The microscope also provides depth, width, volume for every scratch. The volume, force, feed rate and spindle speed were used to calculate the max power, energy and Specific Grinding Energy (SGE). But in this project we only concentrate on the roughness values not the force data.



Figure 12- Keyence VHX-7000

The data recorded in 100µm range of the divot at the lowest point. It is possible to determine the lowest point in the divot if there is no error present in the measurements. At 100µm it provides a more focused region for data collection to focusing on the entire scratch.

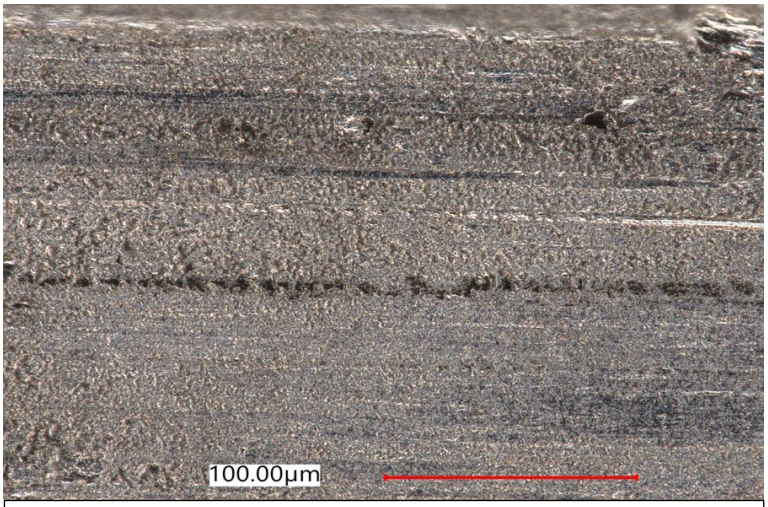


Figure 13- Divot at 1000x

The line roughness values were taken at a magnification of 1000x with each line being 250µm. We need to place 5 lines from the beginning of the scratch to ending of the scratch. First, we draw at the center, one at the top, one at the bottom and two lines bisecting the others. Figure 14 shows the roughness.

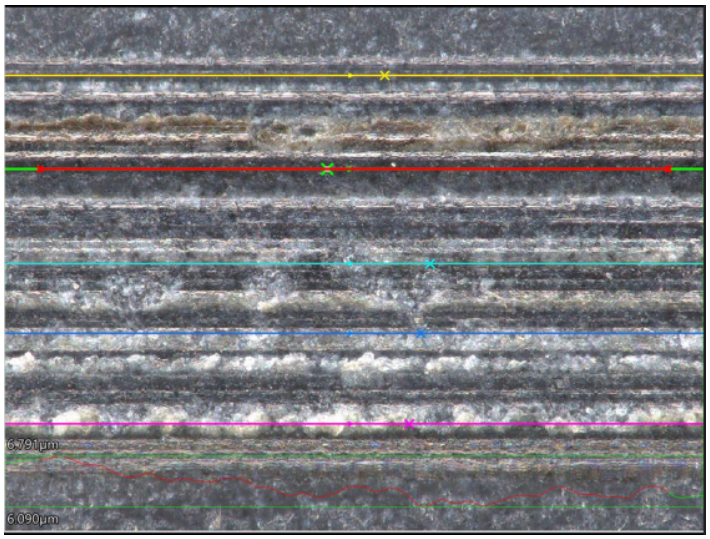


Figure 14- Line Roughness

The volume of the roughness was taken at the width of 250 microns and height is to be adjusted to best fit of the scratch typically between 180-210 microns. We place the rectangular box as needed to view as much as the scratch possible. Figure 15 shows the volume.

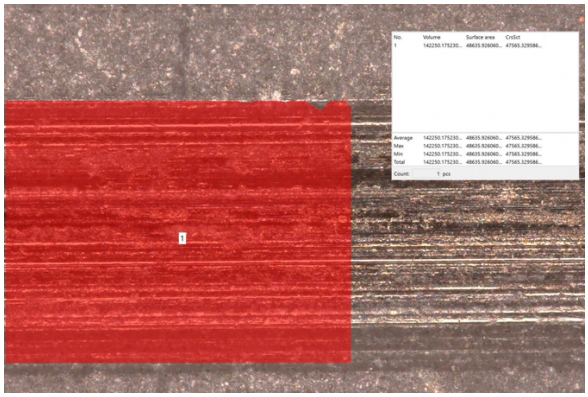


Figure 15- Volume

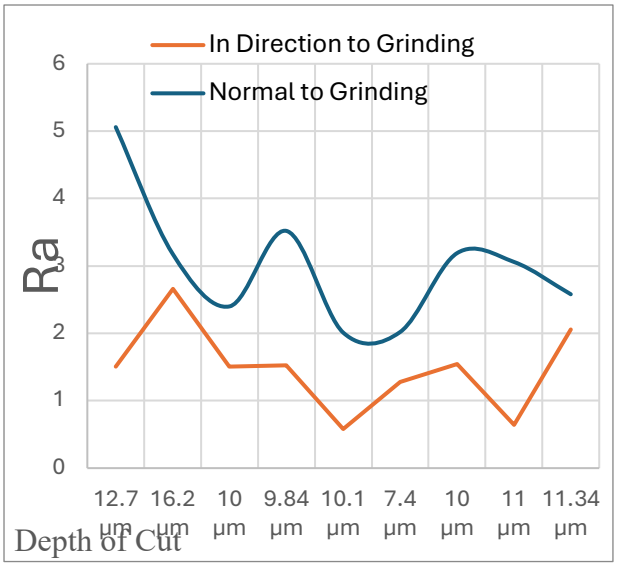
5. Results:

We determine the comparison between 16% and 5% fluid. Below is the data set generated for both fluids. The depth of cut is different for both the fluids.

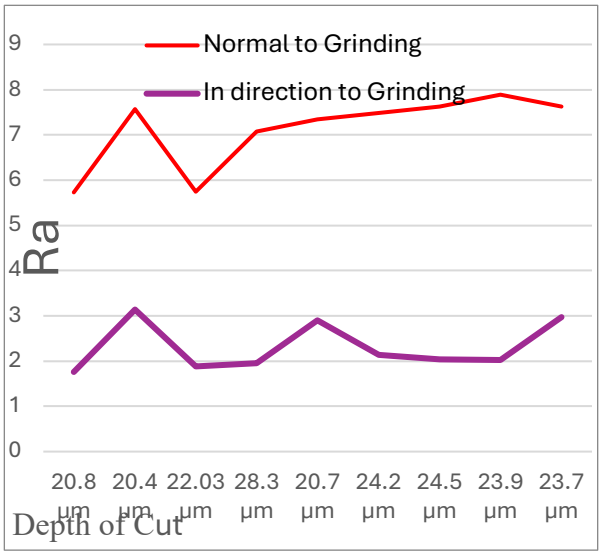
		Normal to Grinding				In the Direction of Grinding			
Concentration	Depth	Ra	RzJIS	Rsk	Rku	Ra	RzJIS	Rsk	Rku
16%	12.7 µm	5.06	6.89	0.03	1.4	1.51	1.66	1.37	7.1
	16.2 µm	3.18	5.24	0.44	2.5	2.66	0	-0.49	3
	10 µm	2.4	0	-0.49	3	1.51	0	-0.14	1.6
	9.84 µm	3.52	3.57	0.16	1.6	1.53	5.65	-0.02	1.8
	10.1 µm	2.01	2.08	0.07	1.9	0.58	0	1.91	7.3
	7.4 µm	2.02	2.42	-0.02	2	1.28	4.11	0.4	2.8
	10 µm	3.19	6.23	-0.32	1.8	1.54	3.03	0.45	2.5
	11 µm	3.06	2.84	0.59	1.9	0.645	0.271	0.564	2.385
	11.34 µm	2.58	4.14	0.269	1.64	2.054	0.693	0.269	1.74

		Normal to Grinding				In the Direction of Grinding			
Concentratio	Depth	Ra	RzJIS	Rsk	Rku	Ra	RzJIS	Rsk	Rku
5%	20.8 µm	5.73	10.19	-0.93	1.64	1.76	0.85	-1.51	2.51
	20.4 µm	7.57	10.95	-0.06	1.4	3.14	0.23	0.11	2.13
	22.03 µm	5.75	6.67	-0.33	1.71	1.88	2.05	0.62	3.17
	28.3 µm	7.07	12.68	-0.47	1.63	1.95	0.63	-0.3	3.24
	20.7 µm	7.35	10.14	0.35	1.56	2.9	1.55	-0.16	2.29
	24.2 µm	7.48	11.86	-0.89	1.44	2.14	1.48	0.59	1.91
	24.5 µm	7.63	11.78	-0.5	1.49	2.04	2.04	1.44	2.69
	23.9 µm	7.89	11.45	-0.08	1.47	2.02	0.01	0.87	2.69
	23.7 µm	7.63	11.78	-0.5	0.23	2.97	0.72	1.49	2.23

The data set we produced is for 4 parameters that is Ra, RzJIS, Rsk, Rku in 2 directions Normal to grinding and in direction of grinding for both the concentration and we did get a graph for parameter to determine which is the best concentration when using the liquids during grinding process.

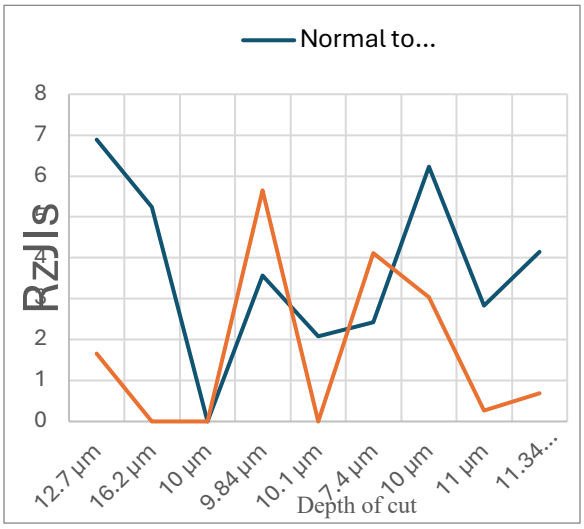


16% Concentration

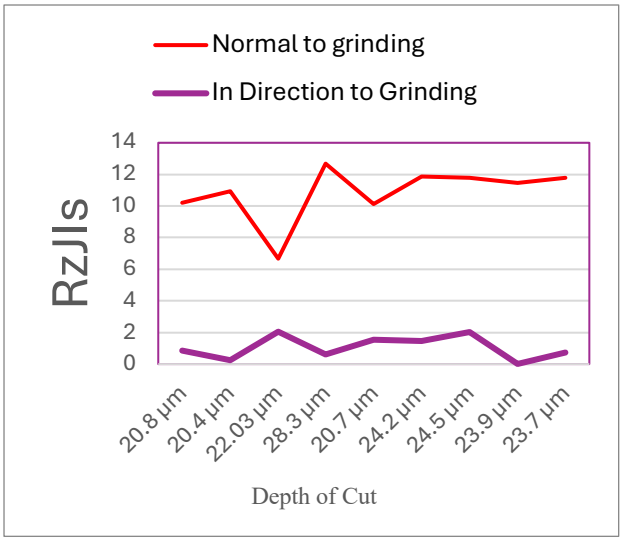


5% Concentration

Ra represents the average deviation of the roughness profile from the mean line over a measured length. It is commonly used to evaluate the overall smoothness of surface. In the above graphs the Ra values were taken from 2 direction, in the direction of grinding and normal to it. The roughness in the direction of grinding is lower considered to the direction normal to it. The depth of cut was not considered a variable parameter and therefore it acts as categories rather than a numerical series. So in the 16% concentration the Ra values indicated smoother surface due to improved lubrication and thermal management.

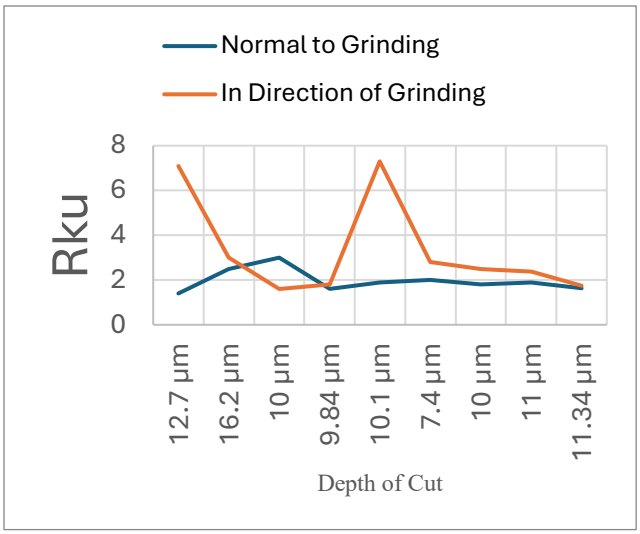


16% Concentration

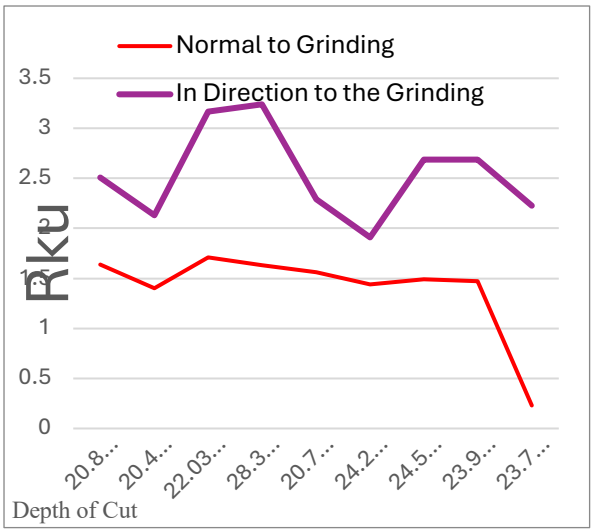


5% Concentration

The RzJIS is the average of 10 points, in which 5 are peak heights and 5 are valley depths. It was extensively used in the Japanese standards hence the letters JIS. The RzJIS values in the normal direction is considerably higher due to the absence of grinding abrasive overlap and single pass.



16% Concentration



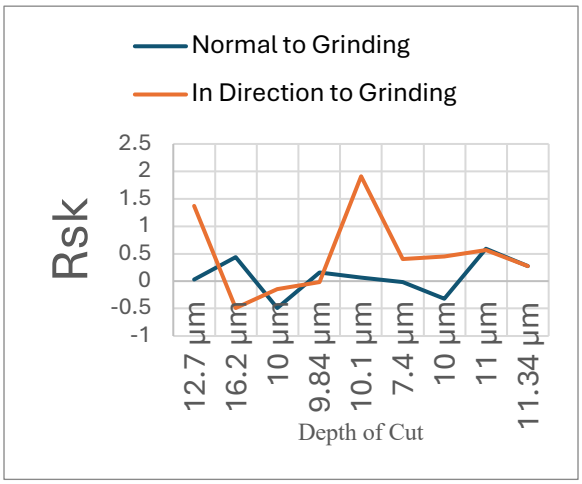
5% Concentration

- Rku describes the sharpness of asperities which is :

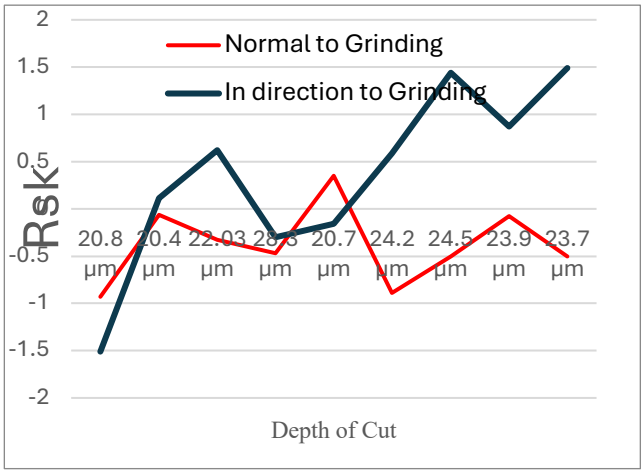
Rku > 3: Indicates sharp peaks or deep valleys.

Rku = 3: Represents a Gaussian distribution.

Rku < 3: Suggests flattened peaks and valleys.



16% Concentration



5% Concentration

The Rsk values are statistical indications of how close the asperities are to the mean line. The negative values of Rsk indicates the presence of more valleys than peaks. The Rsk values are negative normal to the grinding direction.

6. Conclusion:

The findings underscore the potential of sustainable MWFs to address the key challenges in Manufacturing, including carbon emissions and improving cutting quality and tool life. The fluid delivery system played a crucial role in ensuring accurate and consistent testing conditions, contributing to reliable and actionable results. The results indicate that certain advanced MWFs, such as bio based or nanoparticles-enhanced formulations, significantly reduce environmental impacts while maintaining or improving efficiency.

Benefits of 16% concentration:

- Produces smoother and more uniform surfaces (lower RzJIS and stable Rku values).
- Superior cooling and lubrication performance.

5% Coolant Concentration:

- Rougher and less consistent surface quality (higher RzJIS and variable Rku).
- Poor lubrication and thermal management.

The 16% coolant concentration outperforms the 5% concentration in terms of surface quality, stability, and efficiency, making it the better choice.

7. References:

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